

A 16 GB 1024 GB/s HBM3 DRAM with On-Die Error Control Scheme for Enhanced RAS Features

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Abstract

This paper presents the symbol-based On-die ECC (OD-ECC) configuration of High Bandwidth Memory-3 (HBM3) to correct a 16-bit error, bounded by a sub-wordline (WL), and implementation for parallelized data bus inversion (DBI). In addition, the die-to-die integration method for error check and scrub (ECS) mode, and programmable memory built-in self-test (MBIST) design approach for at-speed test are described. The fabricated HBM3 improves the error detection rate by 92.2% with 99.7% fault coverage of OD-ECC logic while achieving 8.0 Gb/s/pin.

Keywords: DRAM, HBM3, RAS, ECC, MBIST, ECS.

Introduction

HBM is the most preferred memory due to its high bandwidth, low power consumption, and large capacity [1]. However, many challenges regarding system reliability, availability, and serviceability (RAS) have been raised because of its industrial usage in automobiles, data centers, and AIs. In order to meet server RAS, recent system-on-chips adopt i) Chipkill, which can correct any errors that occur in a single DRAM device on a DIMM, ii) DDR5-BF, which can correct multi-bit errors by bounding the error patterns in DRAM and applying symbol correction code on the host [2]. However, it is cumbersome to apply Chipkill or DDR5-BF to HBM where the cache line is fetched from a single Core-die (C-die) within the stack [3]. Therefore, single error correction (SEC) has been mainly applied to HBM so far. However, HBM3 is the first JEDEC HBM to apply the symbol based ECC and transparency managed by DRAM to improve multi-bit error controllability for enhanced RAS.

This paper presents a 16 GB HBM3 circuit and practical techniques considering server RAS, while achieving its bandwidth up to 8.0 Gb/s/pin with the 3rd generation of a 10nm DRAM process. The specific contributions of this work are as follows: i) configuration of OD-ECC in HBM3, ii) its implementation considering power, timing, and design-for-test (DFT), and iii) inter-die integration for ECS and instruction-set based programmable MBIST for at-speed test.

Configuration of OD-ECC in proposed HBM3

Fig. 1 (a) shows the chip and bus architecture of HBM3 presented in this work. In C-die, the data-bus line consists of BG_BUS, GBUS, and through-silicon-via (TSV). The BG_BUS is a bank group data-bus line located in the array region. The GBUS is a global data-bus line that delivers data of the BG_BUS to TSV in the DWORD area. The data of BG_BUS, consisting of 256-bit data, 16-bit metadata, and 32 OD-ECC check bits, is divided into two ECC units, and each can correct one symbol (8-bit) by applying (19, 17) Reed-Solomon (RS) code over GF(2⁸), respectively, as shown in

Fig. 1 (b). Therefore, these two ECCs can correct a burst 16-bit error caused by a sub-WL fault, as well as two random single-bit errors (SBEs): one SBE corrected per ECC. HBM2E, on the other hand, can only correct SBEs [4].

In order to provide high error detection rate, 16-bit metadata can be used as error detection code, such as cyclic redundancy check (CRC) code, on the host. Additionally, the error detection rate can be further improved by 92.2% owing to the real-time transparency reported by OD-ECC, in HBM3, referred as error severity (SEV): no error, correctable error, or uncorrectable error as shown in Table I.

OD-ECC with parallelized DBI computation

DBI, a commonly used approach to reduce the power of data-bus, is adopted to BG_BUS, which toggle consumes 22~42% of the power of the C-die. A DBI flag is added to each 8-bit sub-bus unit to indicate whether the sub-bus should be inverted to minimize its toggle. By encoding BG_BUS with the flag, the power of C-die can be reduced by 19% in the maximum toggle case, as shown in Fig. 2 (a). As shown in Fig. 2 (b), by matching the sub-bus (8-bit) unit to the one symbol (8-bit) unit of OD-ECC, hardware logic can be shared, and DBI and OD-ECC can be computed in a parallel manner. This reduces the area overhead as well as the computation time during the read operation.

Since the OD-ECC logic has high test complexity due to the data (304-bit) pattern dependency for computation of check bits (32-bit) or syndrome (32-bit), it requires a lot of overhead to achieve high fault coverage. In order to overcome the test time representing test cost, the OD-ECC logic is isolated optimally from the other data transaction logic by applying non-detectable fault analysis in early design time using the automatic test pattern generation (ATPG). Fig. 3 shows the fault coverage depending on the number of test patterns. Note that, 99.72% of fault coverage can be achieved within 750 patterns.

Implementation of ECS and MBIST

The ECS mode, providing the transparency as described in JEDEC Specification, is applied to HBM3 to support server RAS techniques such as page retirement [3]. Error log registers, implemented in C-die, store the address and the severity of an error. The ECS request from the host is transferred by using the internal serial interface which includes encoded 4-bit command and 4-bit preamble, and the serialized error information delivered to Buffer-die (B-die) accordingly. This internal protocol makes it possible to transmit the burst data with limited TSVs, as shown in Fig. 4 (a), and the timing diagram is illustrated in Fig. 4 (b).

To improve testability and serviceability, MBIST supports at-speed test and programmable instruction-set based pattern

generation in HBM3, while low-frequency (LF) cell-test engine and high-frequency (HF) peripheral-test engine are separated with a few fixed patterns in HBM2E. The programmable MBIST illustrated in Fig. 5, is implemented as follows: i) We adopt globally asynchronous locally synchronous (GALS) with serializing techniques. Thus the pattern generator can operate in LF domain for optimizing power and the area overhead, whereas DRAM cells and internal logic can be tested in HF domain for reflecting real transaction, ii) Programmable instruction-set, configured by users, can generate flexible patterns to enhance test coverage after SiP assembly. Moreover, eFuse also simplifies long instruction-set resulting from the flexibility, and minimizes unnecessary setup process on the host.

Results

Fig. 6 (a) and (b) shows 16 GB HBM3 microphotographs of C-die and B-die. Table II shows the comparison of HBM2E [4] and HBM3. Fig. 6 (c) shows that wide operating VDD ranges from 0.9 V to 1.3 V and the measured tCK at 1.1 V is 0.25 ns which corresponds to 8.0 Gb/s/pin.

References

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TABLE I Error detection rate depending on CRC and real-time SEV.

System ECC	w/o CRC	w/ CRC-8	w/ CRC-16
w/o real-time SEV	-	99.6094%	99.9985%
w/ real-time SEV	92.1875%	99.9695%	99.9999%

TABLE II Comparison of HBM2E [4] and HBM3.

Items	HBM2E	HBM3
Process	2 nd gen of 10nm DRAM	3 rd gen of 10nm DRAM
Cube density	16/8GB (8/4H)	24/16GB (12/8H)
Pin Data Rate	5.0 Gb/s/pin	8.0 Gb/s/pin
Bandwidth	640 GB/s/cube	1024 GB/s/cube
Core-die density	16Gb+2Gb (ECC) + 1.5Gb (OD-ECC)	16Gb+1Gb(metadata) + 2Gb (OD-ECC)
Correction Capability	1-bit error	16-bit error (bounded by sub-WL)
Transparency	Not supported	Real-time SEV/ECS
MBIST	LF-test	United HF-test
Cell and core test	w/ fixed patterns	w/ instruction-set
MBIST	HF-test	based flexible patterns
Peripheral test	w/ fixed patterns	

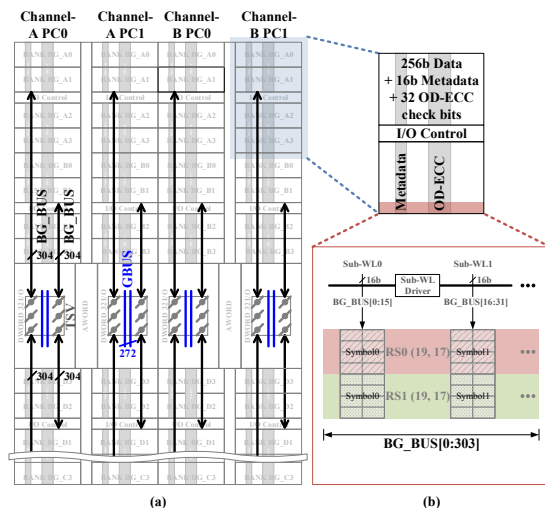


Fig. 1. (a) Chip and bus architecture of HBM3, (b) configuration of OD-ECC.

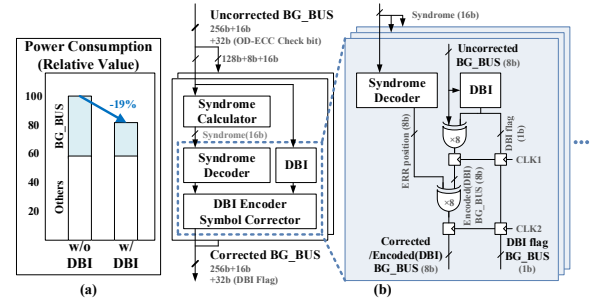


Fig. 2. (a) Power consumption of BG_BUS, (b) block diagram of OD-ECC with parallelized DBI computation.

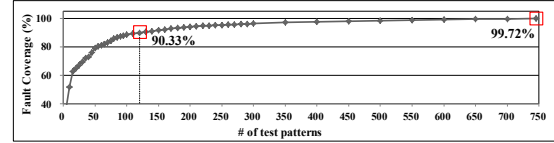


Fig. 3. Fault coverage of OD-ECC depending on the number of test patterns.

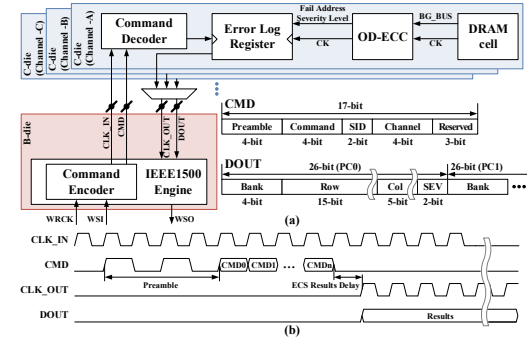


Fig. 4. (a) Block diagram of ECS engine, (b) proposed internal serial interface for die-to-die communication.

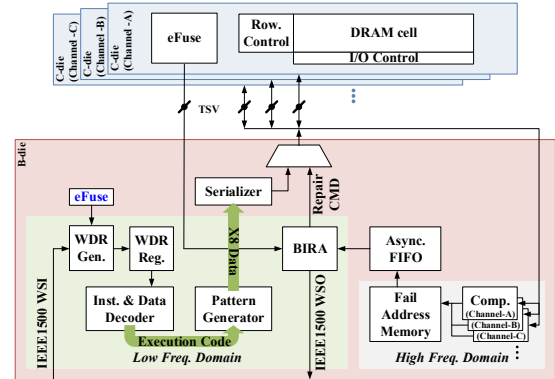


Fig. 5. Block diagram of proposed programmable MBIST.

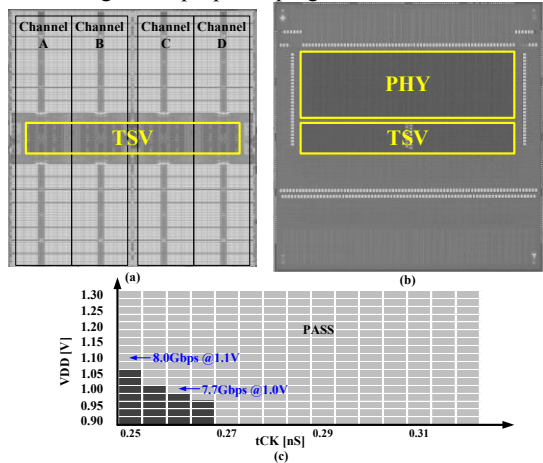


Fig. 6. Chip microphotographs of (a) Core-die and (b) Buffer-die, (c) measured tCK shmoo.